Math 110 — Week 11

Adjoints, self-adjoint and normal operators, the spectral theorem (Notes Ch. 19–20, LADR 7A–7B, HW11)

§1 Facts used from earlier weeks

V and W are finite-dimensional inner product spaces over $\mathbb{F} = \mathbb{R}$ or $\mathbb{C}$, and $n = \dim V$. A bar means complex conjugation; for real numbers it changes nothing.

(T1) **Rules of the inner product** (Week 9). $\langle \lambda u + v, w\rangle = \lambda\langle u, w\rangle + \langle v, w\rangle$, $\langle u, \lambda v + w\rangle = \bar{\lambda}\langle u, v\rangle + \langle u, w\rangle$, $\langle u, v\rangle = \overline{\langle v, u\rangle}$, $\langle v, v\rangle > 0$ for $v \neq 0$, and $\|v\|^2 = \langle v, v\rangle$.

(T2) **Cauchy–Schwarz** (Week 9, Prop. 82). $|\langle u, v\rangle| \leq \|u\|\,\|v\|$.

(T3) **Orthonormal bases** (Week 10, Props. 86–87). In an orthonormal basis $e_1, \ldots, e_n$, $v = \langle v, e_1\rangle e_1 + \cdots + \langle v, e_n\rangle e_n$. $\|a_1 e_1 + \cdots + a_n e_n\|^2 = |a_1|^2 + \cdots + |a_n|^2$ (Prop. 84). An orthonormal list is linearly independent, so n orthonormal vectors form a basis (Prop. 85). δ_{ij} is 1 if $i = j$ and 0 otherwise. Gram–Schmidt turns a basis $v_1, \ldots, v_n$ into an orthonormal basis $e_1, \ldots, e_n$ such that, for every k, $\operatorname{span}(e_1, \ldots, e_k)$ equals $\operatorname{span}(v_1, \ldots, v_k)$; applied to a basis of a subspace, it gives an orthonormal basis of that subspace.

(T4) **Riesz representation** (Week 10, Prop. 88). For every linear functional ψ on V there is exactly one $u \in V$ with $\psi(v) = \langle v, u\rangle$ for all v. In particular, if $\langle v, u\rangle = \langle v, u'\rangle$ for every v, then $u = u'$ (take $v = u - u'$: then $\|u - u'\|^2 = 0$).

(T5) **Orthogonal complements** (Week 10, Props. 89–90). For a subspace U: $V = U \oplus U^\perp$, $\dim U^\perp = n - \dim U$, and $(U^\perp)^\perp = U$.

(T6) **Dual map** (Week 5). For $T \in \mathcal{L}(V, W)$, $T' \in \mathcal{L}(W', V')$ is $T'(\varphi) = \varphi \circ T$.

(T7) **Eigenvalues** (Week 6, Props. 53–54). On a nonzero finite-dimensional complex space every operator has an eigenvalue. Eigenvectors for distinct eigenvalues are linearly independent.

(T8) **Minimal polynomial and diagonalizability** (Week 7). m_T is the monic polynomial of least degree with $m_T(T) = 0$; $q(T) = 0$ exactly when m_T divides q (Prop. 65); the eigenvalues of T are the roots of m_T in $\mathbb{F}$ (Prop. 66); T is diagonalizable exactly when V has a basis of eigenvectors, exactly when $V = E(\lambda_1, T) \oplus \cdots \oplus E(\lambda_m, T)$, exactly when the dimensions of the eigenspaces add up to n (Prop. 62), and exactly when m_T is a product of distinct linear factors (Prop. 68). If T is diagonalizable and U is T-invariant, $T|_U$ is diagonalizable (F3).

(T9) **Real polynomials** (Week 7, F11 step A; uses the fundamental theorem of algebra, cited). Every real polynomial is a product of real linear factors and real quadratics $x^2 + bx + c$ with $b^2 < 4c$.

(T10) **Upper-triangular matrices** (Week 7, Props. 59, 61). For any basis $v_1, \ldots, v_n$, $\mathcal{M}(T)$ is upper-triangular exactly when each $\operatorname{span}(v_1, \ldots, v_k)$ is invariant (Prop. 59). Over $\mathbb{C}$ such a basis exists (Prop. 61).

(T11) **Jordan chains** (Week 8, Prop. 76). Over $\mathbb{C}$, V has a basis made of Jordan chains: lists $v_1, \ldots, v_k$ with $Tv_1 = \lambda v_1$ and $Tv_j = \lambda v_j + v_{j-1}$ for $j \geq 2$. T is diagonalizable exactly when every chain has length 1 (the box after Prop. 76).

(T12) **2×2 matrices** (Week 6). The characteristic polynomial of a 2×2 matrix A is $x^2 - (\operatorname{tr} A)x + \det A$, and its roots are the eigenvalues.

(T13) **Rank–nullity** (Week 3). For $T \in \mathcal{L}(V, W)$, $\dim \operatorname{null} T + \dim \operatorname{range} T = \dim V$; an injective operator on a finite-dimensional space is invertible.

(T14) **Projectors** (Week 10, Prop. 91, Def. 48). If $P^2 = P$, then $V = \operatorname{null} P \oplus \operatorname{range} P$ and P is the identity on $\operatorname{range} P$. The orthogonal projection P_U onto U is the identity on U and 0 on $U^\perp$.

§2 This week

§2.1 The adjoint (Ch. 19)

The question. On $\mathcal{P}_1(\mathbb{R})$ with $\langle f, g\rangle = \int_0^1 f(x)g(x)\,dx$, let D be differentiation. Is there an operator D^* with $\langle Df, g\rangle = \langle f, D^*g\rangle$ for all f, g, and what is $D^*(1)$? The matrix of D in the basis $1, x$ is the matrix below; is D^* given by its transpose?

$$\mathcal{M}(D) = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

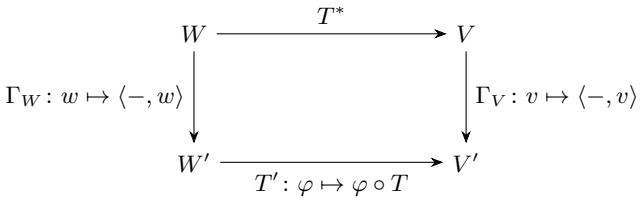
Definition (Def. 49 / LADR 7A)**.** Let $T \in \mathcal{L}(V, W)$. The **adjoint** of T is the map $T^*\colon W \to V$ defined by

$$\langle Tv, w\rangle = \langle v, T^*w\rangle \qquad \text{for all } v \in V,\ w \in W.$$

Other texts write $T^\dagger$.

Why T^* exists and is unique. Fix $w \in W$. The map $v \mapsto \langle Tv, w\rangle$ is a linear functional on V (linear in v by (T1)). By Riesz (T4) there is exactly one vector $u \in V$ with $\langle Tv, w\rangle = \langle v, u\rangle$ for all v; this u is T^*w.

The same thing in the language of dual spaces: write $\Gamma_V(v) = \langle -, v\rangle$ for the map $V \to V'$ of Week 10. The defining equation says $\langle -, T^*w\rangle = \langle -, w\rangle \circ T = T'(\langle -, w\rangle)$, that is, $\Gamma_V(T^*w) = T'(\Gamma_W(w))$. So $T^* = \Gamma_V^{-1} \circ T' \circ \Gamma_W$: the adjoint is the dual map, moved from the dual spaces back to W and V by Riesz. The identification Γ depends on the chosen inner product, so T^* does too. Over $\mathbb{C}$, Γ satisfies $\Gamma(\lambda v) = \bar{\lambda}\,\Gamma(v)$, which is where the conjugates below come from.



Both paths from W to V' agree: $\Gamma_V(T^*w) = T'(\Gamma_W(w))$. Since Γ_V is a bijection, this determines T^*w.

Example 72: the adjoint of differentiation, by solving equations. In the question, write $D^*(1) = ax + b$. The definition with $f = 1$ and with $f = x$ gives two equations:

$$\langle 1, ax + b\rangle = \langle D(1), 1\rangle = 0, \qquad \langle x, ax + b\rangle = \langle D(x), 1\rangle = \langle 1, 1\rangle = 1.$$

The integrals are $\langle 1, ax + b\rangle = \frac{1}{2}a + b$ and $\langle x, ax + b\rangle = \frac{1}{3}a + \frac{1}{2}b$. Solving $\frac{1}{2}a + b = 0$, $\frac{1}{3}a + \frac{1}{2}b = 1$ gives $a = 12$, $b = -6$:

$$D^*(1) = 12x - 6.$$

For $D^*(x) = cx + d$: $\langle 1, cx + d\rangle = \langle D(1), x\rangle = 0$ and $\langle x, cx + d\rangle = \langle D(x), x\rangle = \langle 1, x\rangle = \frac{1}{2}$, so $c = 6$, $d = -3$ and $D^*(x) = 6x - 3$.

Theorem (Prop. 93 / LADR 7.4, 7.9: the matrix of the adjoint is the conjugate transpose)

T^* is linear. If $e_1, \dots, e_n$ is an orthonormal basis of V, $f_1, \dots, f_m$ an orthonormal basis of W, and $\mathcal{M}(T) = (a_{ij})$ in these bases, then $\mathcal{M}(T^*) = (\overline{a_{ji}})$: the conjugate transpose of $\mathcal{M}(T)$.

The idea

Both claims come from moving T across the inner product. For linearity: compare the inner products of $T^*(\lambda w_1 + w_2)$ and of $\lambda T^* w_1 + T^* w_2$ with every v; two vectors with the same inner products against every v are equal (T4). For the matrix: in an orthonormal basis an entry is an inner product, $b_{kj} = \langle T^* f_j, e_k \rangle$, and that inner product can be moved to T.

Proof. 1. T^* **is linear.** For $v \in V$, $w_1, w_2 \in W$, $\lambda \in \mathbb{F}$, by (T1) and the definition,

$$\langle v, T^*(\lambda w_1 + w_2) \rangle = \langle Tv, \lambda w_1 + w_2 \rangle = \bar{\lambda} \langle Tv, w_1 \rangle + \langle Tv, w_2 \rangle = \langle v, \lambda T^* w_1 + T^* w_2 \rangle.$$

This holds for every v, so $T^*(\lambda w_1 + w_2) = \lambda T^* w_1 + T^* w_2$ by (T4).

2. **The matrix.** Column j of $\mathcal{M}(T^*)$ holds the coordinates $b_{1j}, \dots, b_{nj}$ of $T^* f_j$, and by (T3) $b_{kj} = \langle T^* f_j, e_k \rangle$. Moving T across:

$$b_{kj} = \langle T^* f_j, e_k \rangle = \overline{\langle e_k, T^* f_j \rangle} = \overline{\langle Te_k, f_j \rangle} = \overline{a_{jk}},$$

since $a_{jk} = \langle Te_k, f_j \rangle$ is the f_j-coordinate of Te_k (T3).

□

Since D^* is linear, Example 72 gives D^* on every $ax + b$:

$$D^*(ax + b) = a\, D^*(x) + b\, D^*(1) = a(6x - 3) + b(12x - 6) = (3a + 6b)(2x - 1).$$

Example 73: Example 72 again, with an orthonormal basis. $e_1 = 1$ and $e_2 = \sqrt{3}(2x-1)$ are orthonormal for $\langle f, g \rangle = \int_0^1 fg$ ($\|e_2\|^2 = 3\int_0^1 (2x-1)^2 dx = 1$, and $\int_0^1 (2x-1)\, dx = 0$). Since $D(e_1) = 0$ and $D(e_2) = 2\sqrt{3} = 2\sqrt{3}\, e_1$, Prop. 93 gives

$$\mathcal{M}(D) = \begin{bmatrix} 0 & 2\sqrt{3} \\ 0 & 0 \end{bmatrix}, \qquad \mathcal{M}(D^*) = \begin{bmatrix} 0 & 0 \\ 2\sqrt{3} & 0 \end{bmatrix}.$$

So $D^*(e_1) = 2\sqrt{3}\, e_2 = 12x - 6$ and $D^*(e_2) = 0$. Since $x = \frac{1}{2\sqrt{3}} e_2 + \frac{1}{2} e_1$, $D^*(x) = \frac{1}{2} D^*(1) = 6x - 3$, as before.

The question answered. D^* exists, and $D^*(1) = 12x - 6$. The transpose of the matrix in the basis $1, x$ would give $D^*(1) = x$, which is wrong: $1, x$ is not orthonormal ($\langle 1, x \rangle = \frac{1}{2}$), and Prop. 93 needs orthonormal bases.

Proposition (LADR 7.5: rules for adjoints)

For $S, T \in \mathcal{L}(V, W)$, $\lambda \in \mathbb{F}$, $R \in \mathcal{L}(U, V)$ (with U another finite-dimensional inner product space):

$$(S + T)^* = S^* + T^*, \quad (\lambda T)^* = \bar{\lambda} T^*, \quad (T^*)^* = T, \quad I^* = I, \quad (TR)^* = R^* T^*.$$

In particular $(T - \lambda I)^* = T^* - \bar{\lambda} I$ for an operator T; and if T is invertible, so is T^*, with $(T^*)^{-1} = (T^{-1})^*$.

Proof. Each rule is checked the same way: compute $\langle (\text{map})v, w \rangle$, move each factor across, and use the uniqueness in (T4).

1. **Sums.** For all v, w,

$$\langle (S + T)v, w \rangle = \langle Sv, w \rangle + \langle Tv, w \rangle = \langle v, S^* w \rangle + \langle v, T^* w \rangle = \langle v, (S^* + T^*)w \rangle.$$

2. **Scalars.** $\langle \lambda Tv, w \rangle = \lambda \langle v, T^* w \rangle = \langle v, \bar{\lambda} T^* w \rangle$, because a scalar leaves the second slot conjugated (T1).

3. **Adjoint of the adjoint.** Here $T^* \colon W \to V$, so $(T^*)^*$ maps V to W. For $w \in W$, $v \in V$:

$$\langle T^* w, v \rangle = \overline{\langle v, T^* w \rangle} = \overline{\langle Tv, w \rangle} = \langle w, Tv \rangle,$$

so $(T^*)^* v = Tv$ by (T4).

4. **Identity.** $\langle Iv, w\rangle = \langle v, w\rangle = \langle v, Iw\rangle$ for all v, w, so $I^* = I$ by (T4).
5. **Products.** $\langle TRu, w\rangle = \langle Ru, T^*w\rangle = \langle u, R^*T^*w\rangle$, so $(TR)^*w = R^*T^*w$ by (T4): the order reverses.
6. **Shifts.** By the sum, scalar and identity rules, $(T - \lambda I)^* = T^* + (-\lambda I)^* = T^* - \bar{\lambda} I^* = T^* - \bar{\lambda} I$.
7. **Inverses.** Taking adjoints of $TT^{-1} = I$ and $T^{-1}T = I$ with the product rule gives $(T^{-1})^*T^* = I$ and $T^*(T^{-1})^* = I$.

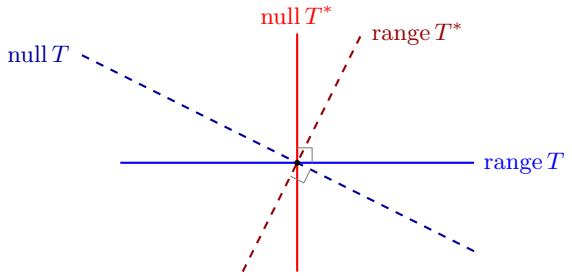
□

Theorem (Prop. 94 / LADR 7.6: null spaces and ranges of the adjoint)
For $T \in \mathcal{L}(V, W)$: (i) $\operatorname{null} T^* = (\operatorname{range} T)^\perp$; (ii) $\operatorname{range} T^* = (\operatorname{null} T)^\perp$.

Proof. 1. **(i).** $w \in \operatorname{null} T^*$ means $T^*w = 0$, which by (T4) is the same as $\langle v, T^*w\rangle = 0$ for every $v \in V$, that is, $\langle Tv, w\rangle = 0$ for every v: w is orthogonal to every vector of $\operatorname{range} T$.

2. **(ii).** Apply (i) to T^* in place of T, using $(T^*)^* = T$: $\operatorname{null} T = (\operatorname{range} T^*)^\perp$. Take orthogonal complements of both sides and use $(U^\perp)^\perp = U$ (T5): $(\operatorname{null} T)^\perp = \operatorname{range} T^*$.

□



$T(x, y) = (x + 2y, 0)$ on $\mathbb{R}^2$, with $T^*(x, y) = (x, 2x)$ (the transpose). $\operatorname{range} T$ (the x-axis) and $\operatorname{null} T^*$ (the y-axis) are perpendicular; so are $\operatorname{null} T = \operatorname{span}(-2, 1)$ and $\operatorname{range} T^* = \operatorname{span}(1, 2)$.

Proposition (LADR 7.7: injectivity of T and surjectivity of T^*)
For $T \in \mathcal{L}(V, W)$: T is injective $\iff$ T^* is surjective; T is surjective $\iff$ T^* is injective; $\dim \operatorname{range} T^* = \dim \operatorname{range} T$; and $\dim \operatorname{null} T^* = \dim \operatorname{null} T + \dim W - \dim V$.

Proof. 1. T is injective exactly when $\operatorname{null} T = \{0\}$, exactly when $(\operatorname{null} T)^\perp = V$ (T5), which by Prop. 94(ii) says $\operatorname{range} T^* = V$.

2. T is surjective exactly when $\operatorname{range} T = W$, exactly when $(\operatorname{range} T)^\perp = \{0\}$ (T5), which by Prop. 94(i) says $\operatorname{null} T^* = \{0\}$.

3. By Prop. 94(ii) and (T5), $\dim \operatorname{range} T^* = n - \dim \operatorname{null} T$, which is $\dim \operatorname{range} T$ by rank–nullity (T13).

4. By Prop. 94(i) and (T5), $\dim \operatorname{null} T^* = \dim W - \dim \operatorname{range} T = \dim W - (\dim V - \dim \operatorname{null} T)$.

□

Proposition (LADR 7A: eigenvalues of T^*)
For an operator T, λ is an eigenvalue of T exactly when $\bar{\lambda}$ is an eigenvalue of T^*.

Proof. By Prop. 94(i) applied to $T - \lambda I$ (whose adjoint is $T^* - \bar{\lambda} I$), $\operatorname{null}(T^* - \bar{\lambda} I) = \operatorname{range}(T - \lambda I)^\perp$. This is nonzero exactly when $T - \lambda I$ is not surjective (T5), which for an operator on a finite-dimensional space means not injective (T13), that is, λ is an eigenvalue of T. □

The eigenvectors can differ: for the matrix B below, $Be_1 = e_1$, but the eigenvectors of B^{T} for 1 are the multiples of e_2.

$$B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \qquad B^{\mathsf{T}} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

Shortcuts

- **Adjoint from a matrix**: in orthonormal bases, conjugate-transpose the matrix. Reason: Prop. 93. Condition: both bases orthonormal. Instance: for $T(z_1, z_2) = (z_1 + iz_2,\, 3z_2)$ on $\mathbb{C}^2$ with the standard inner product, $T^*(z_1, z_2) = (z_1,\, -iz_1 + 3z_2)$.
- **Adjoint for any inner product**: write T^*(basis vector) with unknown coefficients and impose the definition against each basis vector (Example 72); or first find an orthonormal basis (Example 73).
- **Shifting by a scalar**: $(T - \lambda I)^* = T^* - \bar{\lambda} I$. Reason: LADR 7.5. Instance: $(T - iI)^* = T^* + iI$.
- $\operatorname{null} T^*$ **without computing** T^*: it is $(\operatorname{range} T)^{\perp}$ (Prop. 94). Instance: in the figure, $\operatorname{range} T$ is the x-axis, so $\operatorname{null} T^*$ is the y-axis.
- **Injective/surjective**: T injective $\iff$ T^* surjective (LADR 7.7). Instance: $T(x, y) = (x + 2y, 0)$ is not surjective, so T^* is not injective ($T^*(0, 1) = 0$).

Traps

- The transpose rule needs orthonormal bases. In the basis $1, x$ of the question, the transpose gives $D^*(1) = x$, but $D^*(1) = 12x - 6$.
- Over $\mathbb{C}$, scalars come out conjugated: for $T = iI$ on $\mathbb{C}$, $T^* = -iI$, since $\langle iz, w\rangle = iz\bar{w} = \langle z, -iw\rangle$.
- The adjoint of a product reverses the order. On $\mathbb{R}^2$ with S and T below, $(ST)^* = ST$ is the first matrix on the right, but $S^*T^* = S^{\mathsf{T}}T^{\mathsf{T}}$ is the second:

$$S = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}, \quad T = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}; \qquad ST = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}, \qquad S^{\mathsf{T}}T^{\mathsf{T}} = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

- $T \in \mathcal{L}(V, W)$ has $T^* \in \mathcal{L}(W, V)$: the adjoint maps W to V. Only for an operator ($V = W$) can T and T^* be compared or multiplied in either order.

§2.2 Self-adjoint operators (Ch. 19)

The question. The matrix A below equals its own conjugate transpose. Can it have a non-real eigenvalue?

$$A = \begin{bmatrix} 2 & 1 + i \\ 1 - i & 3 \end{bmatrix}$$

Definition (Def. 50 / LADR 7.10)**.** $T \in \mathcal{L}(V)$ is **self-adjoint** if $T = T^*$. Over $\mathbb{R}$ such operators are also called **symmetric**, over $\mathbb{C}$ **Hermitian**. By Prop. 93, T is self-adjoint exactly when its matrix in one (equivalently, every) orthonormal basis equals its conjugate transpose; such a matrix is called Hermitian.

Example 74. The first matrix below is Hermitian; the second equals its transpose but is not Hermitian, because its conjugate transpose has $-i$ off the diagonal.

$$\begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix}$$

Theorem (Prop. 95 / LADR 7.12: eigenvalues of self-adjoint operators are real)
If T is self-adjoint, every eigenvalue of T is real.

The idea

For an eigenvector v, the number $\langle Tv, v\rangle$ equals $\lambda\|v\|^2$. Moving T to the other slot gives the same number, but now λ comes out of the second slot conjugated. So $\lambda = \bar{\lambda}$.

Proof. 1. Let $Tv = \lambda v$ with $v \neq 0$. Using (T1) and $T = T^*$,

$$\lambda\|v\|^2 = \langle \lambda v, v\rangle = \langle Tv, v\rangle = \langle v, Tv\rangle = \langle v, \lambda v\rangle = \bar{\lambda}\|v\|^2.$$

2. Since $\|v\|^2 \neq 0$, $\lambda = \bar{\lambda}$, so λ is real.

□

Over $\mathbb{R}$ this says nothing new (all scalars are real). The converse is false: the matrix below has the real eigenvalue 1 only, but it is not Hermitian.

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

The question answered. No: A is Hermitian, so its eigenvalues are real. Check with (T12): $\operatorname{tr} A = 5$ and $\det A = 6 - (1+i)(1-i) = 4$, so the characteristic polynomial is $\lambda^2 - 5\lambda + 4 = (\lambda - 1)(\lambda - 4)$.

The polarization identity for operators. For any $T \in \mathcal{L}(V)$ write $Q(x) = \langle Tx, x\rangle$. Over $\mathbb{C}$, for all v, w,

$$\langle Tv, w\rangle = \frac{1}{4}\Big(Q(v+w) - Q(v-w) + iQ(v+iw) - iQ(v-iw)\Big).$$

Derivation. Expand by (T1), writing $a = \langle Tv, w\rangle$ and $b = \langle Tw, v\rangle$:

$$Q(v+w) - Q(v-w) = 2\langle Tv, w\rangle + 2\langle Tw, v\rangle = 2a + 2b,$$
$$Q(v+iw) - Q(v-iw) = 2\langle Tv, iw\rangle + 2\langle T(iw), v\rangle = -2ia + 2ib.$$

Multiply the second line by i and add: $(2a + 2b) + (2a - 2b) = 4a$. No hypothesis on T was used. Over $\mathbb{R}$, i is not a scalar, so $v \pm iw$ are not vectors and only the first line is available; and it gives $2a + 2b$; when T is self-adjoint, $b = \langle Tw, v\rangle = \langle w, Tv\rangle = \langle Tv, w\rangle = a$ (real inner product), so $Q(v+w) - Q(v-w) = 4\langle Tv, w\rangle$.

Theorem (Prop. 96 / LADR 7.14: over $\mathbb{C}$, self-adjoint means $\langle Tv, v\rangle$ is always real)
Let V be complex and $T \in \mathcal{L}(V)$. Then T is self-adjoint if and only if $\langle Tv, v\rangle \in \mathbb{R}$ for every $v \in V$.

The idea

One direction is the computation of Prop. 95 without the eigenvector. For the other, the polarization identity recovers every $\langle Tv, w\rangle$ from the values $Q(x) = \langle Tx, x\rangle$; if these are real, the identity with v and w exchanged produces the complex conjugate.

Proof. 1. **Assume T is self-adjoint.** Then $\langle Tv, v\rangle = \langle v, Tv\rangle = \overline{\langle Tv, v\rangle}$ (T1), so $\langle Tv, v\rangle$ is real.

2. **Assume every $Q(x) = \langle Tx, x\rangle$ is real.** Also $Q(\mu x) = \mu\bar{\mu}\, Q(x) = |\mu|^2 Q(x)$, so $Q(-x) = Q(x)$ and $Q(\pm ix) = Q(x)$.

3. Write the polarization identity with v, w exchanged and rewrite each term with step 2: $Q(w+v) = Q(v+w)$, $Q(w-v) = Q(v-w)$, $Q(w+iv) = Q(i(v-iw)) = Q(v-iw)$, $Q(w-iv) = Q(-i(v+iw)) = Q(v+iw)$. So

$$\langle Tw, v\rangle = \frac{1}{4}\Big(Q(v+w) - Q(v-w) + iQ(v-iw) - iQ(v+iw)\Big).$$

4. The Q-values are real, so the right side is the complex conjugate of the polarization formula for $\langle Tv, w\rangle$. Hence $\langle Tw, v\rangle = \overline{\langle Tv, w\rangle}$, that is, $\langle Tv, w\rangle = \overline{\langle Tw, v\rangle} = \langle v, Tw\rangle$ for all v, w. So $T^* = T$ by (T4).

□

So self-adjoint operators play the role of real numbers among operators: for $V = \mathbb{C}$, an operator is multiplication by some z, and it is self-adjoint exactly when $z \in \mathbb{R}$.

Theorem (Prop. 97 / LADR 7.16, 7.13: when $\langle Tv, v\rangle = 0$ for every v)
If $\langle Tv, v\rangle = 0$ for every $v \in V$ and either T is self-adjoint or V is complex, then $T = 0$.

Proof. 1. Here $Q \equiv 0$. Over $\mathbb{C}$ the polarization identity (valid for every T) gives $\langle Tv, w\rangle = 0$ for all v, w. Over $\mathbb{R}$ with T self-adjoint, $4\langle Tv, w\rangle = Q(v+w) - Q(v-w) = 0$.

2. Take $w = Tv$: $\|Tv\|^2 = 0$, so $Tv = 0$ for every v.

□

The hypothesis matters over $\mathbb{R}$: the rotation $T(x, y) = (-y, x)$ has $\langle T(x, y), (x, y)\rangle = -yx + xy = 0$ for every vector, but $T \neq 0$.

Shortcuts

- **Is a matrix Hermitian?** Compare it with its conjugate transpose: the diagonal must be real and $a_{ji} = \overline{a_{ij}}$. Reason: Def. 50 with Prop. 93. Instance: the matrix of the question.
- **Real eigenvalues without computing**: a Hermitian matrix has real eigenvalues (Prop. 95), so its characteristic polynomial has only real roots. Instance: the question.
- **Products**: for self-adjoint S, T, ST is self-adjoint exactly when $ST = TS$. Reason: $(ST)^* = T^*S^* = TS$ (LADR 7.5). Instances: $S = T$ gives that T^2 is self-adjoint; and $\operatorname{diag}(1, 0)$ and $\operatorname{diag}(0, 1)$ commute, so their product 0 is symmetric, while the symmetric matrices below do not commute and their product is not symmetric.
$$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$
- **Sums and real multiples** of self-adjoint operators are self-adjoint. Reason: $(S+T)^* = S^* + T^* = S + T$ and $(cT)^* = \bar{c}T^* = cT$ for real c (LADR 7.5); iT is not, unless $T = 0$, since $(iT)^* = -iT$.
- $\langle Tv, v\rangle$ **test** over $\mathbb{C}$: T is self-adjoint exactly when every $\langle Tv, v\rangle$ is real. Reason: Prop. 96. Instance: for the matrix A of the question and $v = (v_1, v_2)$, $\langle Av, v\rangle = 2|v_1|^2 + 3|v_2|^2 + 2\operatorname{Re}\big((1+i)v_2\bar{v}_1\big)$, which is real.

Traps

- Real eigenvalues do not make an operator self-adjoint: the matrix with rows $(1, 1)$, $(0, 1)$ above.
- Over $\mathbb{C}$, "equals its transpose" is not "Hermitian": the second matrix of Example 74 has eigenvalues $1 \pm i$.
- Over $\mathbb{R}$, $\langle Tv, v\rangle = 0$ for all v does not give $T = 0$ unless T is self-adjoint: the rotation above.

§2.3 Normal operators (Ch. 19)

The question. For the matrices A and B below, compare $\|Ae_1\|$ with $\|A^*e_1\|$, and $\|Be_1\|$ with $\|B^*e_1\|$. For which operators does $\|Tv\| = \|T^*v\|$ hold for every v?

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}, \qquad B = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

Definition (Def. 51 / LADR 7A). $T \in \mathcal{L}(V)$ is **normal** if $TT^* = T^*T$.

Every self-adjoint operator is normal ($TT^* = T^2 = T^*T$). The matrix A of the question is normal but not self-adjoint: AA^* and A^*A are both $2I$, as the computation below shows. B is not normal: BB^* has first row $(2, 1)$, while B^*B has first row $(1, 1)$.

$$AA^* = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}, \qquad A^*A = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}\begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}.$$

Real and imaginary parts. Over $\mathbb{C}$, every operator splits as

$$T = R + iS, \qquad R = \frac{T + T^*}{2}, \quad S = \frac{T - T^*}{2i},$$

and R, S are self-adjoint by LADR 7.5:

$$R^* = \frac{T^* + T}{2} = R, \qquad S^* = \frac{T^* - T}{\overline{2i}} = \frac{T^* - T}{-2i} = S.$$

T is normal exactly when $RS = SR$. *Reason:* multiplying out,

$$(T + T^*)(T - T^*) - (T - T^*)(T + T^*) = 2(T^*T - TT^*), \qquad \text{so} \qquad RS - SR = \frac{T^*T - TT^*}{2i}.$$

These are not the real and imaginary parts of the entries: for the normal matrix below, $R = 0$ and S has entries $\pm i$.

$$\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} + i \begin{bmatrix} 0 & i \\ -i & 0 \end{bmatrix}$$

Theorem (Prop. 98 / LADR 7.20: normal means $\|Tv\| = \|T^*v\|$ for every v)
T is normal if and only if $\|Tv\| = \|T^*v\|$ for every $v \in V$.

The idea

The difference $N = T^*T - TT^*$ measures the failure of normality, and it is self-adjoint. For a self-adjoint operator, $N = 0$ can be tested on the numbers $\langle Nv, v\rangle$ alone (Prop. 97), and $\langle Nv, v\rangle$ is exactly $\|Tv\|^2 - \|T^*v\|^2$.

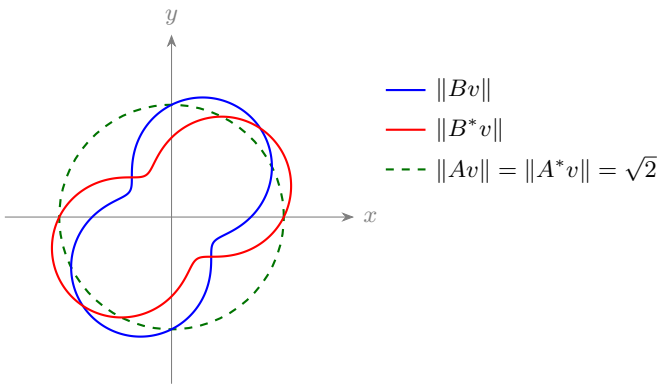
Proof. 1. $N = T^*T - TT^*$ is self-adjoint: by LADR 7.5, $(T^*T)^* = T^*(T^*)^* = T^*T$ and $(TT^*)^* = TT^*$.

2. For every v, moving one factor across each inner product,

$$\langle Nv, v\rangle = \langle T^*Tv, v\rangle - \langle TT^*v, v\rangle = \langle Tv, Tv\rangle - \langle T^*v, T^*v\rangle = \|Tv\|^2 - \|T^*v\|^2.$$

3. T is normal exactly when $N = 0$. By Prop. 97 (N is self-adjoint), $N = 0$ exactly when $\langle Nv, v\rangle = 0$ for every v, which by step 2 says $\|Tv\| = \|T^*v\|$ for every v.

□



In each direction $v = (\cos t, \sin t)$, the distance from 0 to a curve is the length of the image of v. For the normal A both images have length $\sqrt{2}$ (one circle). For B the two curves differ, so B is not normal.

The question answered. $Ae_1 = (1, 1)$ and $A^*e_1 = (1, -1)$ both have length $\sqrt{2}$; $Be_1 = (1, 0)$ has length 1 but $B^*e_1 = (1, 1)$ has length $\sqrt{2}$. By Prop. 98, $\|Tv\| = \|T^*v\|$ for every v holds exactly for the normal operators; A is one, B is not.

Theorem (Prop. 99 / LADR 7.21: null spaces, ranges and eigenvectors of a normal operator)
Let T be normal. Then $\operatorname{null} T^* = \operatorname{null} T$ and $\operatorname{range} T^* = \operatorname{range} T$, so $(\operatorname{null} T)^\perp = \operatorname{range} T$ and $V = \operatorname{null} T \oplus \operatorname{range} T$. Moreover $T - \lambda I$ is normal for every λ, and v is an eigenvector of T for λ exactly when v is an eigenvector of T^* for $\bar{\lambda}$.

Proof. 1. **Null spaces.** By Prop. 98, $\|T^*v\| = 0$ exactly when $\|Tv\| = 0$.

2. **Ranges.** By Prop. 94(ii), $\operatorname{range} T^* = (\operatorname{null} T)^\perp$, and by step 1 this is $(\operatorname{null} T^*)^\perp$. By Prop. 94(i), $\operatorname{null} T^* = (\operatorname{range} T)^\perp$, so $(\operatorname{null} T^*)^\perp = \operatorname{range} T$ by (T5). Hence $\operatorname{range} T^* = \operatorname{range} T$.

3. **Direct sum.** By step 2, $(\operatorname{null} T)^\perp = \operatorname{range} T^* = \operatorname{range} T$, so $V = \operatorname{null} T \oplus (\operatorname{null} T)^\perp = \operatorname{null} T \oplus \operatorname{range} T$ (T5).

4. $T - \lambda I$ **is normal.** Its adjoint is $T^* - \bar{\lambda} I$ (LADR 7.5). Both products expand to the same expression, because $TT^* = T^*T$:
$$(T - \lambda I)(T^* - \bar{\lambda} I) = TT^* - \bar{\lambda} T - \lambda T^* + |\lambda|^2 I = (T^* - \bar{\lambda} I)(T - \lambda I).$$

5. **Eigenvectors.** Apply step 1 to the normal operator $T - \lambda I$: $\operatorname{null}(T - \lambda I) = \operatorname{null}(T^* - \bar{\lambda} I)$.

□

The direct sum can fail without normality: the matrix N with rows $(0, 1)$, $(0, 0)$ has $\operatorname{null} N = \operatorname{range} N = \operatorname{span}(e_1)$.

Proposition (LADR 7.22: orthogonal eigenvectors for normal operators)
If T is normal, $Tv = \lambda v$ and $Tw = \mu w$ with $\lambda \neq \mu$, then $\langle v, w \rangle = 0$.

Proof. 1. By Prop. 99, $T^*w = \bar{\mu} w$.

2. Then $\lambda \langle v, w \rangle = \langle Tv, w \rangle = \langle v, T^*w \rangle = \langle v, \bar{\mu} w \rangle = \mu \langle v, w \rangle$ (T1), so $(\lambda - \mu)\langle v, w \rangle = 0$, and $\langle v, w \rangle = 0$ since $\lambda \neq \mu$.

□

Self-adjoint operators are normal, so this applies to them. Without normality it fails: B with rows $(1, 1)$, $(0, 2)$ has eigenvectors $(1, 0)$ for 1 and $(1, 1)$ for 2, which are not orthogonal.

Shortcuts

- **Test normality**: compute AA^* and A^*A and compare. Reason: Def. 51 with Prop. 93. Instance: A above.
- **Eigenvectors of T^* from those of T**: for normal T, the same vectors with conjugate eigenvalues. Reason: Prop. 99. Instance: A has the eigenvector $(1, -i)$ for $1 + i$, as the computation below checks, so $A^*(1, -i) = (1 - i)(1, -i)$.
$$A \begin{bmatrix} 1 \\ -i \end{bmatrix} = \begin{bmatrix} 1 + i \\ 1 - i \end{bmatrix} = (1 + i) \begin{bmatrix} 1 \\ -i \end{bmatrix}$$
- **Always normal**: self-adjoint operators ($TT^* = T^2 = T^*T$); operators with $T^* = -T$ ($TT^* = -T^2 = T^*T$); operators with $T^*T = I$, such as the rotations and reflections of $\mathbb{R}^n$. Reason for the last: T^* is a left inverse of T, so T is injective, hence invertible (T13), and then $T^* = T^{-1}$ is also a right inverse: $TT^* = I = T^*T$. Instances: the reflection $(x, y) \mapsto (y, x)$ and the rotation $(x, y) \mapsto (-y, x)$.
- **Null space of T^***: for normal T it equals $\operatorname{null} T$; no need to compute T^*. Reason: Prop. 99. Instance: the symmetric matrix with all four entries 1 has null $= \operatorname{span}(1, -1)$, for T and for T^*.

Traps

- Normal is weaker than self-adjoint: A above is normal but $A^* \neq A$, and its eigenvalues $1 \pm i$ are not real.

- A sum of normal operators need not be normal. The first matrix below is symmetric and the second is a rotation, so both are normal; call their sum N:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} = N.$$

  $Ne_1 = (0, 2)$ has length 2, but $N^{\mathsf{T}} e_1 = 0$. So $\|Ne_1\| \neq \|N^* e_1\|$, and N is not normal (Prop. 98).
- $\operatorname{null} T^* = \operatorname{null} T$ can fail without normality: for the matrix with rows $(0, 1)$, $(0, 0)$, the null space is $\operatorname{span}(e_1)$ and the null space of its transpose is $\operatorname{span}(e_2)$.

§2.4 The real spectral theorem (Ch. 20)

The question. The matrix A below is symmetric. Is there an orthonormal basis of $\mathbb{R}^2$ made of eigenvectors of A? What does A do to the unit circle?

$$A = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

Throughout this subsection and the next, $V \neq \{0\}$.

Definition. $T \in \mathcal{L}(V)$ is **orthogonally diagonalizable** if V has an orthonormal basis consisting of eigenvectors of T.

Proposition (LADR 7.26: invertible quadratic expressions)
Let $T \in \mathcal{L}(V)$ be self-adjoint and $b, c \in \mathbb{R}$ with $b^2 < 4c$. Then $T^2 + bT + cI$ is invertible.

The idea

For a real number t, $t^2 + bt + c > 0$ when $b^2 < 4c$. For $v \neq 0$, the number $\langle (T^2 + bT + cI)v, v\rangle$ equals $\|Tv\|^2 + b\langle Tv, v\rangle + c\|v\|^2$. Cauchy–Schwarz bounds the middle term below by $-|b|\,\|Tv\|\,\|v\|$, so the number is at least $\|v\|^2(t^2 - |b|t + c)$ with $t = \|Tv\|/\|v\|$, and $t^2 - |b|t + c > 0$ because $(-|b|)^2 < 4c$. A positive number is not 0, so $(T^2 + bT + cI)v \neq 0$.

Proof. 1. Let $v \neq 0$. Since T is self-adjoint, $\langle T^2 v, v\rangle = \langle Tv, Tv\rangle = \|Tv\|^2$, so

$$\langle (T^2 + bT + cI)v, v\rangle = \|Tv\|^2 + b\langle Tv, v\rangle + c\|v\|^2.$$

2. By Cauchy–Schwarz (T2), $b\langle Tv, v\rangle \geq -|b|\,\|Tv\|\,\|v\|$. Completing the square,

$$\|Tv\|^2 + b\langle Tv, v\rangle + c\|v\|^2 \geq \left(\|Tv\| - \frac{|b|}{2}\|v\|\right)^2 + \left(c - \frac{b^2}{4}\right)\|v\|^2 > 0,$$

since the square is ≥ 0, $c - b^2/4 > 0$, and $\|v\|^2 > 0$.

3. So $(T^2 + bT + cI)v \neq 0$ for every $v \neq 0$: the operator is injective, hence invertible (T13).

□

Theorem (Prop. 100 / LADR 7.27: a self-adjoint operator has an eigenvalue)
Every self-adjoint $T \in \mathcal{L}(V)$ has an eigenvalue (over $\mathbb{R}$ as well as over $\mathbb{C}$).

The idea

Over $\mathbb{C}$ this is (T7). Over $\mathbb{R}$, the eigenvalues are the real roots of m_T (T8). If there were none, m_T would be a product of quadratics $x^2 + bx + c$ with $b^2 < 4c$ (T9); each of these, evaluated at T, is invertible by LADR 7.26, so $m_T(T)$ would be invertible instead of 0.

Proof. 1. If $\mathbb{F} = \mathbb{C}$, (T7) gives an eigenvalue. Let $\mathbb{F} = \mathbb{R}$, and suppose m_T has no real root. By (T9), $m_T = p_1 \cdots p_m$ with each $p_k = x^2 + b_k x + c_k$ and $b_k^2 < 4c_k$.

2. By LADR 7.26 each $p_k(T)$ is invertible, so $m_T(T) = p_1(T)\cdots p_m(T)$ is a product of invertible operators and is invertible. But $m_T(T) = 0$ and $V \neq \{0\}$: a contradiction. So m_T has a real root, which is an eigenvalue (T8).

□

Theorem (Prop. 101 / LADR 7B.19: complements of invariant subspaces are invariant)
Let T be self-adjoint and U a T-invariant subspace. Then $U^\perp$ is T-invariant, and $T|_U \in \mathcal{L}(U)$ and $T|_{U^\perp} \in \mathcal{L}(U^\perp)$ are self-adjoint (with the restricted inner products).

Proof. 1. **$U^\perp$ is invariant.** Let $v \in U^\perp$ and $u \in U$. Then $Tu \in U$, so $\langle Tv, u\rangle = \langle v, Tu\rangle = 0$. Thus Tv is orthogonal to every $u \in U$: $Tv \in U^\perp$.

2. **$T|_U$ is self-adjoint.** For $u, u' \in U$, $\langle T|_U u, u'\rangle = \langle Tu, u'\rangle = \langle u, Tu'\rangle = \langle u, T|_U u'\rangle$.

3. **$T|_{U^\perp}$ is self-adjoint.** By step 1, T maps $U^\perp$ to itself. For $x, x' \in U^\perp$,

$$\langle T|_{U^\perp} x, x'\rangle = \langle Tx, x'\rangle = \langle x, Tx'\rangle = \langle x, T|_{U^\perp} x'\rangle.$$

□

So for a self-adjoint T, every invariant subspace U has an invariant complement, namely $U^\perp$. A non-diagonalizable operator can fail this: for the matrix B with rows $(1,1)$, $(0,1)$, the only eigenvectors are the multiples of e_1, so $\operatorname{span}(e_1)$ is the only invariant line, and it has no invariant complement.

Theorem (Prop. 102 / LADR 7.29: the real spectral theorem)
(i) Every self-adjoint operator is orthogonally diagonalizable (over $\mathbb{R}$ or $\mathbb{C}$). (ii) If V is real, every orthogonally diagonalizable operator is self-adjoint. So over $\mathbb{R}$: T is self-adjoint $\iff$ T is orthogonally diagonalizable.

The idea

Take one unit eigenvector v_1 (Prop. 100). Its line is invariant, so its orthogonal complement is invariant too (Prop. 101), and T restricted there is again self-adjoint, on a space of one dimension less. By induction that complement has an orthonormal basis of eigenvectors, and adding v_1 gives one for V. For (ii), test the definition of self-adjoint on the eigenvector basis, where everything is a number $\lambda_i\delta_{ij}$; the eigenvalues must be real for the scalar to come out of the second slot unchanged.

Proof. 1. **(i), induction on $n = \dim V$.** For $n = 1$, any unit vector is an eigenvector and forms an orthonormal basis.

2. For $n > 1$, take an eigenvector (Prop. 100) and scale it to $\|v_1\| = 1$. $U = \operatorname{span}(v_1)$ is invariant ($T(cv_1) = c\lambda v_1$), so $U^\perp$ is invariant and $T|_{U^\perp}$ is self-adjoint (Prop. 101), with $\dim U^\perp = n - 1$ (T5).

3. By induction $U^\perp$ has an orthonormal basis $v_2, \ldots, v_n$ of eigenvectors of $T|_{U^\perp}$, which are eigenvectors of T. Each is orthogonal to v_1, so $v_1, \ldots, v_n$ is an orthonormal list of n vectors, a basis of V.

4. **(ii).** Let $v_1, \ldots, v_n$ be an orthonormal basis with $Tv_i = \lambda_i v_i$, $\lambda_i \in \mathbb{R}$. For all i, j,

$$\langle Tv_i, v_j\rangle = \lambda_i\delta_{ij} = \lambda_j\delta_{ij} = \langle v_i, \lambda_j v_j\rangle = \langle v_i, Tv_j\rangle,$$

where $\lambda_i\delta_{ij} = \lambda_j\delta_{ij}$ because both are 0 unless $i = j$, and $\langle v_i, \lambda_j v_j\rangle = \lambda_j\langle v_i, v_j\rangle$ because λ_j is real.

5. By linearity in each slot, $\langle Tv, w\rangle = \langle v, Tw\rangle$ for all v, w, so $T = T^*$.

□

Proposition (LADR 7.27: the minimal polynomial of a self-adjoint operator)
If T is self-adjoint, then $m_T = (x - \lambda_1)\cdots(x - \lambda_m)$, where $\lambda_1, \ldots, \lambda_m$ are the distinct eigenvalues of T, all real.

Proof. 1. By Prop. 102(i), T is diagonalizable, so m_T is a product of distinct linear factors (T8).

2. The roots of m_T are the eigenvalues of T (T8), which are real (Prop. 95).

□

Instance: for the matrix A of the question, $m_A = (x-4)(x-2)$.

By LADR 7.22, eigenvectors of a self-adjoint operator for distinct eigenvalues are orthogonal.

In matrix form. A real square matrix Q whose columns are orthonormal satisfies $Q^{\mathsf{T}}Q = I$ (entry (i,j) of $Q^{\mathsf{T}}Q$ is the dot product of columns i and j); such a Q is an **orthogonal matrix**, and $Q^{-1} = Q^{\mathsf{T}}$. For a real symmetric matrix A, put the orthonormal eigenvectors of Prop. 102 into the columns of Q and the eigenvalues into a diagonal D. Then $AQ = QD$, because column j of AQ is $Au_j = \lambda_j u_j$, which is column j of QD. Multiplying on the right by $Q^{-1} = Q^{\mathsf{T}}$,

$$A = QDQ^{\mathsf{T}}, \qquad A^m = QD^mQ^{\mathsf{T}} \quad (m \geq 1).$$

Orthogonal diagonalization of a real symmetric matrix.

1. Find the eigenvalues (they are real) and a basis of each eigenspace.
2. Inside each eigenspace, apply Gram–Schmidt (T3). Different eigenspaces are already orthogonal (LADR 7.22).
3. Collect the unit vectors as the columns of Q; then $A = QDQ^{\mathsf{T}}$.

Worked example: a repeated eigenvalue. For the matrix A below, $A - I$ has all entries 1, so it has rank 1 and $\dim E(1,A) = 3 - 1 = 2$ (T13); and $A(1,1,1) = 4(1,1,1)$, so $\dim E(4,A) \geq 1$. Eigenspaces for distinct eigenvalues form a direct sum (T7), so their dimensions add up to at most 3. Hence $\dim E(4,A) = 1$, there is no other eigenvalue, and the eigenvalues are $4, 1, 1$. By LADR 7.27, $m_A = (x-4)(x-1)$, although the eigenvalue 1 is repeated; indeed $A - I$ is the all-ones matrix J, $J^2 = 3J$, and $(A-4I)(A-I) = (J-3I)J = 0$.

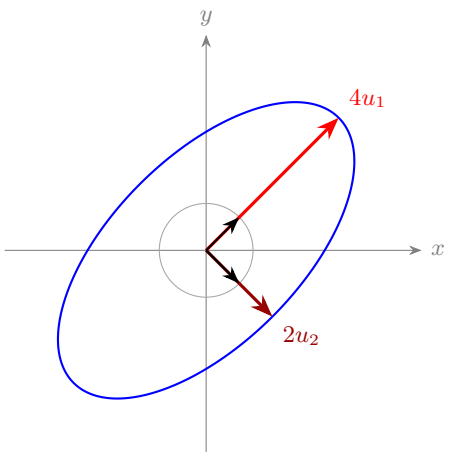
$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$$

$E(1,A)$ is the plane $x + y + z = 0$, with basis $(1,-1,0)$, $(1,0,-1)$. These are not orthogonal (dot product 1), so apply Gram–Schmidt: $e_2 = \frac{1}{\sqrt{2}}(1,-1,0)$, and

$$w_3 = (1,0,-1) - \langle (1,0,-1), e_2 \rangle e_2 = (1,0,-1) - \tfrac{1}{2}(1,-1,0) = \tfrac{1}{2}(1,1,-2), \qquad e_3 = \tfrac{1}{\sqrt{6}}(1,1,-2).$$

With $e_1 = \frac{1}{\sqrt{3}}(1,1,1)$, which is orthogonal to the plane,

$$Q = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & -1/\sqrt{2} & 1/\sqrt{6} \\ 1/\sqrt{3} & 0 & -2/\sqrt{6} \end{bmatrix}, \qquad D = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad A = QDQ^{\mathsf{T}}.$$



A maps the unit circle (gray, with the unit eigenvectors in black) to an ellipse (blue) whose axes point along the orthonormal eigenvectors $u_1 = \frac{1}{\sqrt{2}}(1,1)$ and $u_2 = \frac{1}{\sqrt{2}}(1,-1)$, with half-lengths 4 and 2.

The question answered. Yes: A is symmetric, so Prop. 102 applies. $A(1,1) = 4(1,1)$ and $A(1,-1) = 2(1,-1)$, and $u_1 = \frac{1}{\sqrt{2}}(1,1)$, $u_2 = \frac{1}{\sqrt{2}}(1,-1)$ are orthonormal. Writing $v = au_1 + bu_2$ with $a^2 + b^2 = 1$, $Av = 4au_1 + 2bu_2$: the circle becomes the ellipse in the figure.

Where this is used (Notes Ch. 20). For a reversible Markov chain, the transition operator is self-adjoint for a weighted inner product built from the stationary distribution. So its eigenvalues are real and it has an orthonormal eigenbasis: the eigenvalue 1 is the equilibrium, and the other eigenvalues describe how fast deviations from equilibrium decay: each step multiplies the component along an eigenvector by its eigenvalue, so the largest absolute value ρ among the eigenvalues other than 1 sets the rate, and $1 - \rho$ is called the **spectral gap**. The adjacency matrix of an undirected graph is symmetric, which is the starting point of spectral graph theory. The Hessian matrix of a smooth function is symmetric.

Shortcuts

- 2×2 **symmetric**: once one eigenvector (a, b) is found, the other is $(-b, a)$. Reason: eigenvectors for distinct eigenvalues are orthogonal, and in $\mathbb{R}^2$ the vectors orthogonal to (a, b) are the multiples of $(-b, a)$. Condition: distinct eigenvalues (a repeated eigenvalue of a symmetric 2×2 matrix means $A = \lambda I$). Instance: the question.
- **Distinct eigenvalues**: the eigenvectors are already orthogonal; only normalize. Gram–Schmidt is needed only inside an eigenspace of dimension ≥ 2. Reason: LADR 7.22. Instance: the question, where $(1, 1)$ and $(1, -1)$ are only divided by $\sqrt{2}$.
- **Last eigenspace in $\mathbb{R}^3$**: if the eigenvalues are λ (twice) and μ, then $E(\lambda, A) = E(\mu, A)^\perp$. Reason: $E(\lambda, A) \subseteq E(\mu, A)^\perp$ by LADR 7.22, and both have dimension 2 (Prop. 102, T5). Instance: in the worked example, $E(1, A)$ is the plane orthogonal to $(1, 1, 1)$.
- **Powers and polynomials**: $A^m = QD^mQ^\mathsf{T}$ and $p(A) = Qp(D)Q^\mathsf{T}$. Reason: in $(QDQ^\mathsf{T})^m$ each inner $Q^\mathsf{T}Q$ is I. Instance: for the question, with u_1, u_2 as columns,

$$A^m = \frac{4^m}{2}\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} + \frac{2^m}{2}\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}.$$

- **Check**: multiply A by each column of Q. Instance: in the worked example, $A(1, 1, -2) = (1, 1, -2)$.

Traps

- Eigenvectors for the same eigenvalue need not be orthogonal: $(1, -1, 0)$ and $(1, 0, -1)$ in the worked example have dot product 1. Orthogonalize inside the eigenspace.
- Part (ii) fails over $\mathbb{C}$: $T = iI$ on $\mathbb{C}$ is orthogonally diagonalizable (every unit vector is an eigenvector), but $T^* = -iI \neq T$.
- A complex symmetric matrix is not Hermitian and need not have real eigenvalues: the second matrix of Example 74.

§2.5 The complex spectral theorem (Ch. 20)

The question. The matrix below is normal but not self-adjoint, so Prop. 102 does not apply. Is there still an orthonormal basis of $\mathbb{C}^2$ made of eigenvectors?

$$A = \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

In this subsection V is complex.

Theorem (Prop. 103: normal operators are diagonalizable)
If $T \in \mathcal{L}(V)$ is normal, then T is diagonalizable.

The notes give five independent proofs. Each uses a different earlier tool, and each shows a different reason why normality rules out non-diagonal behaviour.

The idea

Proof 1: not diagonalizable means m_T has a repeated root (T8), and then there is an operator $q(T) \neq 0$ with $q(T)^2 = 0$. For a normal operator S, $\|S(Sv)\| = \|S^*(Sv)\|$, and $\langle S^*Sv, v\rangle = \|Sv\|^2$; so $S^2 = 0$ forces $S = 0$. Polynomials in a normal T are normal, so $q(T) = 0$, contradicting the minimality of m_T.

Proof 1 (minimal polynomial). 1. Suppose m_T has a repeated root: $m_T = (x-\lambda)^2 r$ for some $\lambda \in \mathbb{C}$ and polynomial r. Put $q = (x-\lambda)r$. Then $q^2 = m_T \cdot r$ is divisible by m_T, so $q(T)^2 = 0$ (T8); and $\deg q < \deg m_T$, so $q(T) \neq 0$.

2. $S = q(T)$ **is normal.** By LADR 7.5, $q(T)^* = \bar{q}(T^*)$, where $\bar{q}$ has the conjugated coefficients. Since $TT^* = T^*T$, every power of T commutes with every power of T^*, so $q(T)$ commutes with $\bar{q}(T^*)$.

3. For every v, apply Prop. 98 to S and the vector Sv: $0 = \|S(Sv)\|^2 = \|S^*Sv\|^2$, so $S^*S = 0$.

4. Then $0 = \langle S^*Sv, v\rangle = \langle Sv, Sv\rangle = \|Sv\|^2$ for every v, so $S = q(T) = 0$, contradicting step 1.

5. So m_T has no repeated root. Over $\mathbb{C}$ it splits into linear factors (FTA, cited), which are therefore distinct, and T is diagonalizable (T8).

□

The idea

Proof 2: write $T = R + iS$ with commuting self-adjoint R and S. Each is diagonalizable by the real spectral theorem, and because they commute, each eigenspace of S is invariant under R, so one basis diagonalizes both. A common eigenvector of R and S is an eigenvector of T.

Proof 2 (real and imaginary parts). 1. Write $T = R + iS$ with R, S self-adjoint and $RS = SR$ (normal operators section). By Prop. 102(i), R and S are diagonalizable.

2. Each eigenspace $E(\mu, S)$ is R-invariant: if $Sv = \mu v$, then $S(Rv) = RSv = \mu Rv$.

3. $R|_{E(\mu,S)}$ is diagonalizable (T8, restriction), so each $E(\mu, S)$ has a basis of eigenvectors of R; these are also eigenvectors of S (for μ).

4. V is the direct sum of the eigenspaces of S (T8), so these bases together form a basis of V. Each vector v in it has $Rv = \rho v$ and $Sv = \mu v$, so $Tv = (\rho + i\mu)v$: a basis of eigenvectors of T.

□

The idea

Proof 3: split V into the eigenspace $\text{null}(T - \lambda I)$ for one eigenvalue and its orthogonal complement. For normal T that complement is $\text{range}(T - \lambda I)$ (Prop. 99), which is invariant under T and T^*, so T restricted to it is normal on a smaller space, and induction applies.

Proof 3 (induction on $\dim V$*).* 1. For $n = 1$ there is nothing to prove. Let $n > 1$, assume the result in dimensions below n, and let λ be an eigenvalue (T7). Put $U = \text{range}(T - \lambda I)$. If $U = \{0\}$, then $T = \lambda I$ and every basis of V consists of eigenvectors; so assume $U \neq \{0\}$. Since $T - \lambda I$ is not injective, it is not surjective (T13), so $1 \leq \dim U < n$.

2. U is invariant under T and under T^*: for $u = (T - \lambda I)v$, $Tu = (T - \lambda I)Tv \in U$ and, since T^* commutes with T, $T^*u = (T - \lambda I)T^*v \in U$.

3. $T|_U$ is normal: for $u, u' \in U$, $\langle Tu, u'\rangle = \langle u, T^*u'\rangle$ with $T^*u' \in U$, so the adjoint of $T|_U$ in U is $T^*|_U$, and these commute. By induction U has a basis of eigenvectors of T.

4. $U^\perp = \text{range}(T-\lambda I)^\perp = \text{null}\big((T-\lambda I)^*\big) = \text{null}(T^* - \bar{\lambda} I)$ by Prop. 94(i) and LADR 7.5, and this is $\text{null}(T-\lambda I)$ by Prop. 99. Every nonzero vector there is an eigenvector for λ.
5. $V = U \oplus U^\perp$ (T5): the two bases together are a basis of eigenvectors of T.

□

> **The idea**
>
> Proof 4: a Jordan chain of length at least 2 puts its first vector both in $\text{null}(T-\lambda I)$ and in $\text{range}(T-\lambda I)$. For normal T these two subspaces are orthogonal (Prop. 99), so that vector is orthogonal to itself and is 0.

Proof 4 (Jordan chains). 1. Suppose a Jordan chain $v_1, \dots, v_k$ for λ has $k \geq 2$ (T11). Then $v_1 = (T-\lambda I)v_2$ lies in $\text{range}(T-\lambda I)$, and $(T-\lambda I)v_1 = 0$, so $v_1 \in \text{null}(T-\lambda I)$.

2. $T - \lambda I$ is normal (Prop. 99), so $\text{null}(T-\lambda I) = \text{range}(T-\lambda I)^\perp$ (Prop. 99 with Prop. 94). Thus v_1 is orthogonal to itself: $v_1 = 0$, which a chain vector cannot be.
3. So all chains have length 1, and T is diagonalizable (T11).

□

> **Proposition** (LADR 6.38: Schur's theorem)
>
> Every operator on a complex V has an upper-triangular matrix in some *orthonormal* basis.

Proof. Take a basis $v_1, \dots, v_n$ with each $\text{span}(v_1, \dots, v_k)$ invariant (T10). Gram–Schmidt gives an orthonormal basis with $\text{span}(e_1, \dots, e_k) = \text{span}(v_1, \dots, v_k)$ for every k (T3), so the same spans are invariant, and the matrix in $e_1, \dots, e_n$ is upper-triangular (T10). □

> **The idea**
>
> Proof 5: in an orthonormal basis where $\mathcal{M}(T)$ is upper-triangular, $\|Te_1\|$ is the length of column 1, which holds only a_{11}, and $\|T^*e_1\|$ is the length of row 1. Normality makes the two equal (Prop. 98), so the rest of row 1 is 0. Repeat row by row.

Proof 5 (Schur). 1. Take an orthonormal basis $e_1, \dots, e_n$ with $\mathcal{M}(T) = (a_{ij})$ upper-triangular (Schur). Then $Te_k = a_{1k}e_1 + \cdots + a_{nk}e_n$, so by Prop. 84 (T3), $\|Te_k\|^2 = |a_{1k}|^2 + \cdots + |a_{nk}|^2$, the sum over column k. By Prop. 93, $T^*e_k = \overline{a_{k1}}e_1 + \cdots + \overline{a_{kn}}e_n$, so $\|T^*e_k\|^2 = |a_{k1}|^2 + \cdots + |a_{kn}|^2$, the sum over row k.

2. **Row 1.** Column 1 holds only a_{11}. Prop. 98 gives

$$|a_{11}|^2 = \|Te_1\|^2 = \|T^*e_1\|^2 = |a_{11}|^2 + |a_{12}|^2 + \cdots + |a_{1n}|^2,$$

   so $a_{12} = \cdots = a_{1n} = 0$.
3. **Row k, by induction.** Suppose rows $1, \dots, k-1$ are 0 off the diagonal. Then column k holds only a_{kk}: the entries a_{jk} with $j < k$ lie off the diagonal in rows $j < k$, and those with $j > k$ lie below the diagonal. Prop. 98 gives $|a_{kk}|^2 = |a_{kk}|^2 + |a_{k,k+1}|^2 + \cdots + |a_{kn}|^2$, so row k is 0 off the diagonal too.
4. So $\mathcal{M}(T)$ is diagonal, and $e_1, \dots, e_n$ are eigenvectors.

□

Proof 5 gives more than Prop. 103: the eigenvector basis it finds is orthonormal. The next theorem gets this in general.

> **Theorem** (Prop. 104 / LADR 7.31: the complex spectral theorem)
>
> For $T \in \mathcal{L}(V)$, V complex: T is normal $\iff$ T is orthogonally diagonalizable.

Proof. 1. **Assume T is normal.** It is diagonalizable (Prop. 103), so V is the direct sum of its eigenspaces (T8).

2. Eigenspaces for distinct eigenvalues are orthogonal (LADR 7.22). The notes argue directly: if $Tv = \lambda v$, $Tw = \mu w$ and $\lambda \neq \mu$, then $(T - \lambda I)w = (\mu - \lambda)w$, so

$$w = (\mu - \lambda)^{-1}(T - \lambda I)w \in \operatorname{range}(T - \lambda I) = \operatorname{null}(T - \lambda I)^{\perp}$$

   by Prop. 99 for the normal $T - \lambda I$; since $v \in \operatorname{null}(T - \lambda I)$, $\langle v, w \rangle = 0$.

3. Choose an orthonormal basis of each eigenspace (T3). By step 2 the union is orthonormal, and by step 1 it has n vectors: an orthonormal basis of eigenvectors.

4. **Assume $v_1, \ldots, v_n$ is an orthonormal basis with $Tv_i = \lambda_i v_i$.** By Prop. 93, the matrix of T^* in this basis is the conjugate transpose of the diagonal matrix of T, so $T^* v_i = \bar{\lambda}_i v_i$. Then

$$T^*Tv_i = \lambda_i \bar{\lambda}_i v_i = |\lambda_i|^2 v_i, \qquad TT^*v_i = \bar{\lambda}_i \lambda_i v_i = |\lambda_i|^2 v_i.$$

5. T^*T and TT^* agree on a basis, so $T^*T = TT^*$.

□

In matrix form. A complex square matrix U with orthonormal columns has $U^*U = I$ (entry (i, j) is $\langle u_j, u_i \rangle$), so $U^{-1} = U^*$; such a U is called **unitary**. A normal matrix is $A = UDU^*$ with U unitary and D diagonal.

Worked example: the matrix of the question. A is normal (shown in the normal-operators section). $\operatorname{tr} A = 2$ and $\det A = 2$, so the eigenvalues are the roots of $\lambda^2 - 2\lambda + 2$: $1 \pm i$. For $1 + i$, the first row of $A - (1 + i)I$ is $(-i, -1)$, so $(1, -i)$ is an eigenvector. For $1 - i$, the vector $(1, i)$ is an eigenvector, by the computation on the left below. The two are orthogonal (LADR 7.22), as the computation on the right confirms:

$$A \begin{bmatrix} 1 \\ i \end{bmatrix} = \begin{bmatrix} 1 - i \\ 1 + i \end{bmatrix} = (1 - i) \begin{bmatrix} 1 \\ i \end{bmatrix}, \qquad \langle (1, -i), (1, i) \rangle = 1 \cdot 1 + (-i) \cdot \bar{i} = 1 - 1 = 0.$$

Normalizing,

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ -i & i \end{bmatrix}, \qquad D = \begin{bmatrix} 1 + i & 0 \\ 0 & 1 - i \end{bmatrix}, \qquad A = UDU^*.$$

The question answered. Yes: A is normal, so Prop. 104 applies; the orthonormal eigenvectors are $\frac{1}{\sqrt{2}}(1, -i)$ and $\frac{1}{\sqrt{2}}(1, i)$. Over $\mathbb{R}$ there is none, since A has no real eigenvalue.

Shortcuts

- **Orthogonally diagonalizable over $\mathbb{C}$?** Check $AA^* = A^*A$; no eigenvectors needed. Reason: Prop. 104. Instance: the question.
- **Second eigenvector of a normal** 2×2: orthogonal to the first. Reason: LADR 7.22. Condition: distinct eigenvalues. Instance: the worked example.
- **Polynomials of a normal operator**: in an orthonormal eigenbasis, $p(T)$ is diagonal with entries $p(\lambda_i)$, so $p(T)$ is normal; and T is self-adjoint exactly when all λ_i are real. Reason: Prop. 104 in both directions; T^* is diagonal with entries $\bar{\lambda}_i$ (Prop. 93). Instance: for A of the question, A^2 has eigenvalues $(1 \pm i)^2 = \pm 2i$ and is the normal matrix with rows $(0, -2)$, $(2, 0)$.
- **Inverse of a unitary matrix**: $U^{-1} = U^*$, no row reduction. Reason: $U^*U = I$ and U is square (T13). Instance: for U of the worked example, U^{-1} has rows $\frac{1}{\sqrt{2}}(1, i)$ and $\frac{1}{\sqrt{2}}(1, -i)$.

Traps

- Diagonalizable is weaker than orthogonally diagonalizable. The matrix with rows $(1, 1)$, $(0, 2)$ has distinct eigenvalues $1, 2$, but its eigenvectors $(1, 0)$ and $(1, 1)$ are not orthogonal, and it is not normal.
- Over $\mathbb{R}$, normal operators need not be diagonalizable: the rotation $(x, y) \mapsto (-y, x)$ is normal and has no real eigenvalue.

- Over $\mathbb{C}$, orthogonal eigenvectors use the complex inner product: $(1,-i)$ and $(1,i)$ are orthogonal, but the sum of products without conjugation, $1+(-i)(i)=2$, is not 0.

§3 Problem set

HW11 P1. Let $T\colon \mathbb{F}^n \to \mathbb{F}^n$ be $T(z_1,\dots,z_n)=(0,z_1,\dots,z_{n-1})$, where $\mathbb{F}$ is $\mathbb{R}$ or $\mathbb{C}$. Find T^*.
Hint: Prop. 93 in the standard (orthonormal) basis.

HW11 P2. Let $T\colon V\to W$ be linear between finite-dimensional inner product spaces. Prove that T is injective if and only if T^* is surjective.
Hint: Prop. 94(ii) and $(\operatorname{null} T)^{\perp}=V \iff \operatorname{null} T=\{0\}$.

HW11 P3. Let $T\colon \mathbb{F}^2\to\mathbb{F}^2$ be the quarter-turn $T(x,y)=(-y,x)$. (a) Find T^*. (b) Prove that T is normal but not self-adjoint. (c) Find all eigenvalues of T when $\mathbb{F}=\mathbb{R}$ and when $\mathbb{F}=\mathbb{C}$.
Hint: Prop. 93; compute TT^* and T^*T; the characteristic polynomial λ^2+1.

HW11 P4. Let P be a projector ($P^2=P$) on a finite-dimensional inner product space. Prove that the following are equivalent: (a) P is self-adjoint; (b) P is normal; (c) P is the orthogonal projection onto $\operatorname{range} P$.
Hint: (b)⇒(c) by Prop. 99: $\operatorname{null} P=(\operatorname{range} P)^{\perp}$; for (c)⇒(a) write $v=u+u'$, $w=x+x'$ along $\operatorname{range} P\oplus(\operatorname{range} P)^{\perp}$.

HW11 P5. Let T be self-adjoint and U a T-invariant subspace. Prove that $U^{\perp}$ is T-invariant.
Hint: Prop. 101, step 1.

HW11 P6. For the matrix A below: (a) find an orthonormal basis of $\mathbb{R}^2$ of eigenvectors of A; (b) find an orthogonal Q and a diagonal D with $A=QDQ^{\mathsf{T}}$; (c) find a formula for A^m, $m\ge 1$.

$$A=\begin{bmatrix}2 & 1\\ 1 & 2\end{bmatrix}$$

Hint: the real spectral theorem and the 2×2 symmetric shortcut; $A^m=QD^mQ^{\mathsf{T}}$.

HW11 P7 (optional). Let T be normal on a finite-dimensional complex inner product space and p a polynomial. Use the complex spectral theorem to prove that $p(T)$ is normal.
Hint: Prop. 104 both ways: in an orthonormal eigenbasis of T, $p(T)$ is diagonal.

O1 (adjoint for a weighted inner product). On $\mathbb{R}^2$ with $\langle x,y\rangle=2x_1y_1+x_2y_2$, let $T(x_1,x_2)=(x_2,0)$. Find T^*, and compare its matrix with the transpose of $\mathcal{M}(T)$ in the standard basis.
Hint: the method of Example 72: write $T^*(e_1)$ and $T^*(e_2)$ with unknown coefficients and impose $\langle Tv,w\rangle=\langle v,T^*w\rangle$ for $v=e_1,e_2$; or rescale e_1 to get an orthonormal basis and use Prop. 93.

O2 (eigenspaces of T^*). Let $T\in\mathcal{L}(V)$ and $\lambda\in\mathbb{F}$. Prove that $\dim E(\bar\lambda,T^*)=\dim E(\lambda,T)$.
Hint: LADR 7.7 for the operator $T-\lambda I$, whose adjoint is $T^*-\bar\lambda I$.

O3 (a normal complex symmetric matrix). For the matrix M below on $\mathbb{C}^2$: is M self-adjoint? normal? Find an orthonormal basis of eigenvectors, if there is one.

$$M=\begin{bmatrix}1 & i\\ i & 1\end{bmatrix}$$

Hint: compare M with M^*, then compute MM^* and M^*M; for the eigenvectors use (T12), and LADR 7.22 for orthogonality.

O4 (real eigenvalues). Let T be normal on a complex V. Prove that T is self-adjoint if and only if every eigenvalue of T is real.
Hint: one direction is Prop. 95. For the other, use an orthonormal eigenbasis (Prop. 104) and Prop. 93.

O5 (LADR 7B). Let T be normal on a complex V with $T^9 = T^8$. Prove that T is self-adjoint and $T^2 = T$.
Hint: in an orthonormal eigenbasis (Prop. 104), $\lambda^9 = \lambda^8$ for each eigenvalue, so $\lambda \in \{0, 1\}$; then use O4.

O6 (nilpotent and self-adjoint). Let T be self-adjoint with $T^2 = 0$. Prove that $T = 0$.
Hint: $\|Tv\|^2 = \langle T^2 v, v\rangle$ when T is self-adjoint.